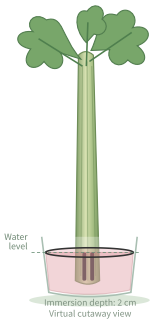
Supported practice





After some time

What the school gardening group found

The gardening group puts celery with leaves in red water. They observe it before and after some time has passed.

Look at the two pictures on the left and record the changes you see. The group wants to know: do light and leaves affect water transport?

Main investigation question

When other conditions are the same, how does light intensity (光強度) affect the upward speed of the red water trace in celery with leaves?

This diagram shows an imaginary cutaway so you can see the water trace inside. The celery used in the experiment has not been cut lengthwise.

Your first observations

Describe visible changes in the trace; keep explanations separate from observations.

Evaluate a hypothesis, not original prediction

Evaluate “Celery with a larger leaf area is better.” Can this statement be tested directly as written? (yes / no) Identify the factor and replace “better” with a measurable result. This is evaluation after learning, not an original prediction.

Identify the variables

For the light comparison using A, B and C, tick one column for each factor.

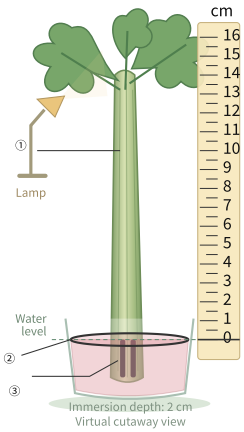
Factor	Independent variable	Dependent variable	Controlled variables
Light intensity (光強度)	☐	☐	☐
Height / average upward speed of the red water trace	☐	☐	☐
Temperature	☐	☐	☐
Humidity (濕度) and air movement	☐	☐	☐
Red water concentration and depth of immersion (浸入深度)	☐	☐	☐
Total leaf area of the setups with leaves	☐	☐	☐
Length and thickness of the celery stalks	☐	☐	☐

Read specified setups; do not design from scratch

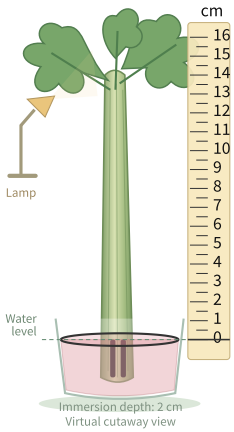
Fixed conditions: 25°C, the same humidity (濕度) and airflow, dye concentration, 2 cm depth of immersion (浸入深度), and similar petiole (葉柄) length and thickness. A, B and C have the same leaf area. B and D have the same relevant conditions except leaf presence.

0 min

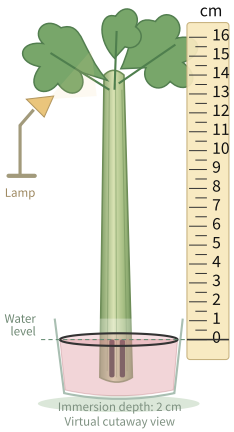
Setup A • With leaves
10 cm • 1200 lux



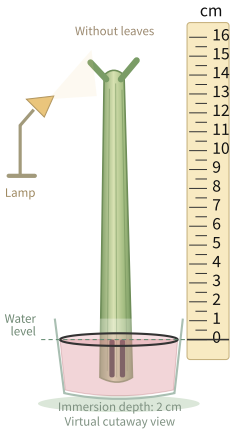
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux



① points to the long stalk, ② to the liquid surface, ③ to the immersed part. Complete: plant part and water-carrying tissue for ①; ruler reading at ②; immersion depth for ③.

For light intensity (光強度) use (A/B/C / A/D); for leaf presence use (B/D / A/D). Why is A/D not a fair comparison of leaf presence?

What assumptions does this investigation make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all the assumptions that apply. You may choose more than one.

- ☐ Changing the lamp distance does not make the setups have different temperatures.
- ☐ Setups A, B and C have the same total leaf area.
- ☐ Leaf area does not affect the results, so it does not need to be kept the same.
- ☐ Humidity (濕度) and air movement are the same for all setups.

Changing lamp distance may change temperature. Why monitor and keep all setups at 25°C?

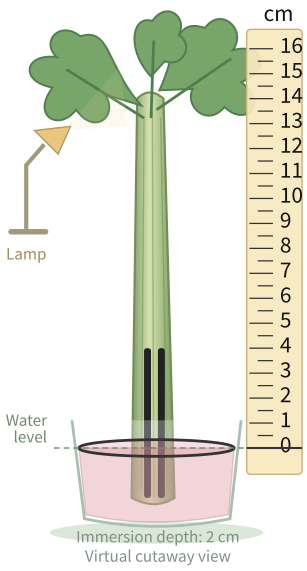
Independent application: read, calculate and analyse

Read 0 min in B2 and 10, 20 and 30 min in B3–B4; fill the B4 table. You may use methods in A; do not read C yet.

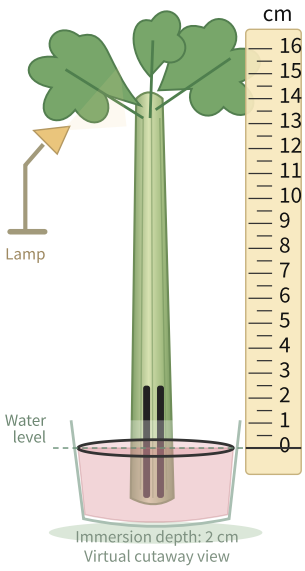
Images use the same teaching model as the VL, not real measurements. Zero stays at the liquid surface; each small division is 0.5 cm. Read the rulers yourself; no answers are listed beside them.

10 min

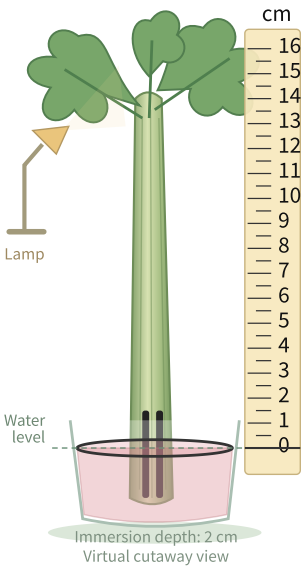
Setup A • With leaves
10 cm • 1200 lux



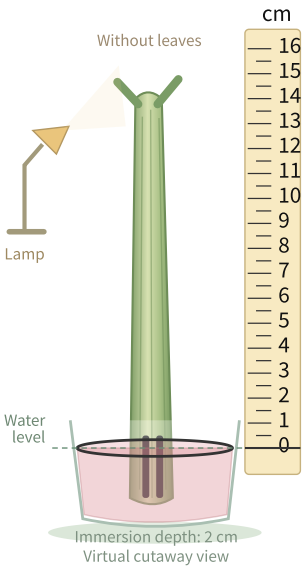
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux

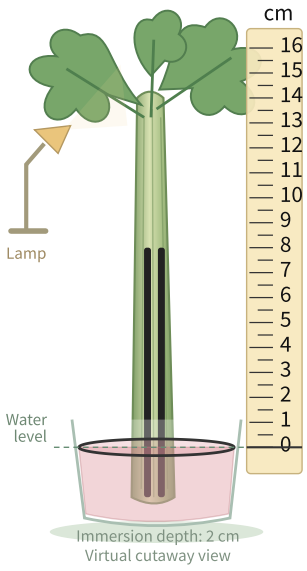


Setup D • Without leaves
20 cm • 300 lux

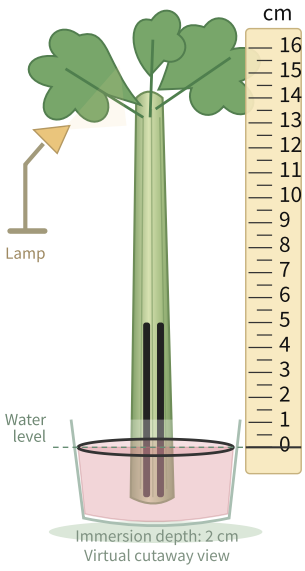


20 min

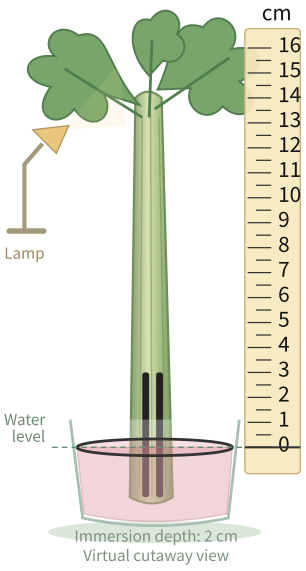
Setup A • With leaves
10 cm • 1200 lux



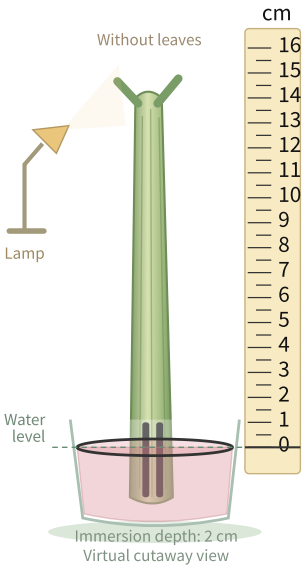
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



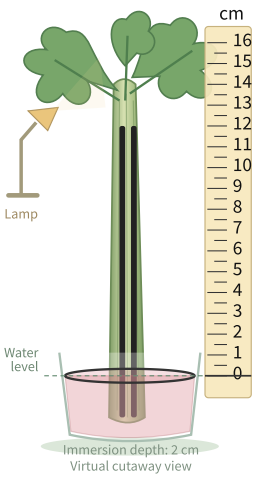
Setup D • Without leaves
20 cm • 300 lux



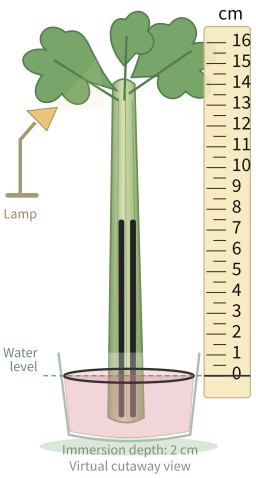
Keep first answers; write corrections in C after checking, without erasing first answers.

30 min

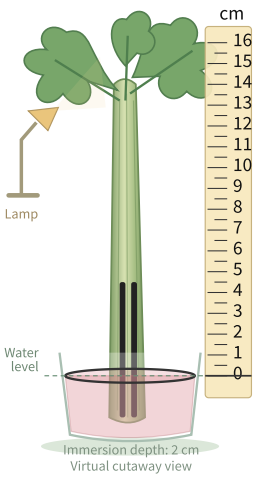
Setup A • With leaves
10 cm • 1200 lux



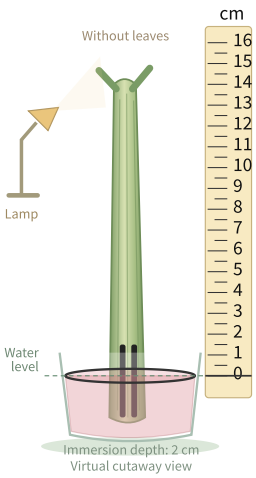
Setup B • With leaves
20 cm • 300 lux



Setup C • With leaves
30 cm • 133 lux



Setup D • Without leaves
20 cm • 300 lux



Your measurement record

min	A (cm)	B (cm)	C (cm)	D (cm)
0				
10				
20				
30				

Keep first answers; write corrections in C after checking, without erasing first answers.

Calculate the average upward speed

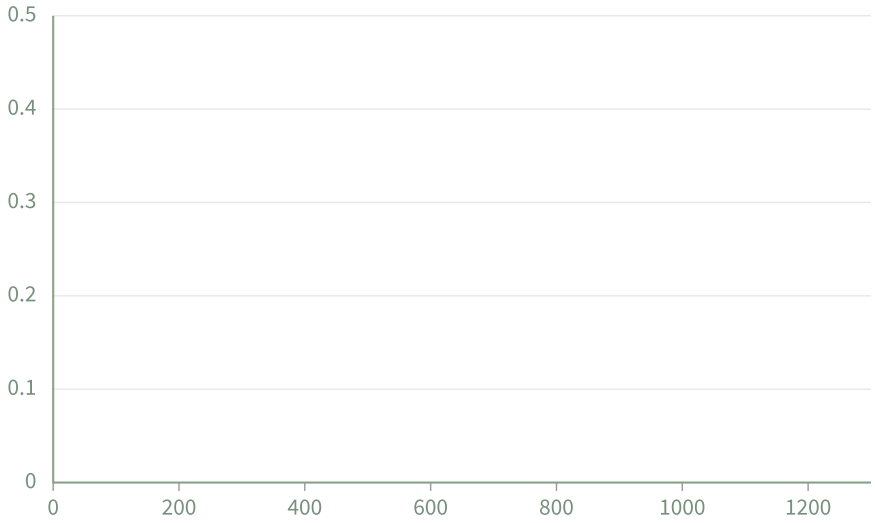
Average upward speed = (height at 30 min − height at 0 min) ÷ 30. Use two decimal places and cm/min.

Setup	Average upward speed (cm/min)
A	
B	
C	
D	

Light intensity (光強度) and the upward speed of the red water trace

Plot four points using your calculations. X: light intensity (光強度) (lux); Y: average upward speed (cm/min). Join A, B and C in light-intensity order to show a reasonable trend. Mark D separately; do not join it into that trend.

Y: Average upward speed of the red water trace (cm/min)



X: Light intensity (光強度) (lux)

Use your graph to reach a conclusion

Circle one answer in each set of brackets.

1. How is light intensity (光強度) related to the upward speed of the water trace?

As light intensity (光強度) increases, the average upward speed of the red water trace usually will (increase / decrease / show no clear change / show no consistent trend).

2. What is the role of leaves?

At the same light distance of 20 cm, the average upward speed of the water trace in Setup B (with leaves), compared with Setup D (without leaves), is (faster / slower / similar).

This comparison supports the idea that leaves (help water move upwards / stop water moving upwards / have no effect on water moving upwards).

3. What can this experiment tell us?

We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?(The red water trace gives an indirect clue about water uptake and upward water transport. Its speed is not the rate of water loss from leaves, or the transpiration rate (蒸騰速率). / The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率). / If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).)

Use data from A, B, C and B/D to support the conclusions.

Can three light-intensity points give exact speeds for all untested values? (yes / no)

Keep first answers; write corrections in C after checking, without erasing first answers.